



Department of Electrical Engineering
University of Moratuwa
B.Sc. Eng. Semester 3



EE3024 - Digital Signal Processing
Semester 03

Final Project
THE ADVANCED LIGHT INTENSITY INDICATOR

Feature 01 – Filtering out unnecessary frequencies

Introduction to the Filtering Requirement

Before any digital processing is carried out, the LDR output must be filtered so that the system measures only the true ambient light changes. In real lighting environments, the LDR voltage can carry unwanted interferences. If this noise is not removed, the system might experience the following interruptions.

Seven-segment display starts to flicker

Rapid binary transitions at the comparator stage

Incorrect averaging at averaging stage (feature 4)

False triggering of “stability detection” at stabilizer stage (feature 3)

Therefore, a dedicated **analog filter** must be placed between the LDR voltage divider and the ADC stages.

Reasons for Design Decisions

1. Why a low-pass filter?

The ambient light level changes; slowly over many seconds with no meaningful components above a few Hertz. Meanwhile, the interference lies at **50–100 Hz**, which is significantly higher than the frequencies associated with natural light changes.

Hence;

A **low-pass filter** can preserve genuine light variations and also frequencies above 10 Hz must be attenuated by **at least 20 dB**.

Feature	Value / Decision	Assumption
Cut-off frequency	10 Hz	The LDR has a relatively slow response time (20–100 ms), which limits its ability to detect meaningful light-level changes to roughly 10 Hz or below. Natural lighting changes occur very slowly and are almost always under 10 Hz, while higher frequencies mainly correspond to electrical noise (50–100 Hz).
Stopband frequency	50 Hz	The dominant interference in indoor lighting systems comes from AC mains flicker at 50–100 Hz.
Attenuation	20 dB	It is sufficient to reduce the 50 Hz noise to a level where it no longer affects the LDR’s output reading.

2. Why a sallen-key 2nd order filter with butter-worth approximation?

Provides a sharper roll-off than 1st-order RC filters

Easy to implement using a single op-amp

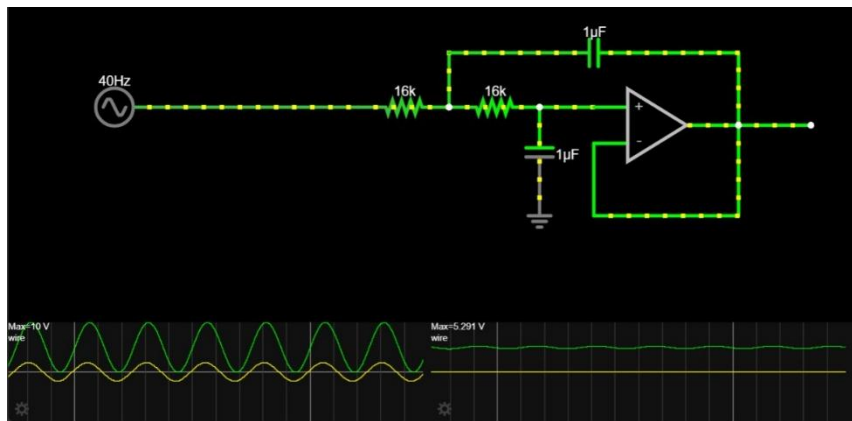
Very stable and well-documented

Requires only 2 resistors, 2 capacitors, and 1 op-amp

Provides a smooth (maximally flat) response when Butterworth parameters are used

Works with the 10 Hz cut-off requirement

Falstad Simulation



Mathematical Design for the Filter

- For a unity-gain Sallen-Key filter: $f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$
- And for a butter-worth : $Q = \frac{1}{\sqrt{2}}$ hence $\frac{R_1}{R_2} = \frac{C_2}{C_1}$
- According to the available specifications in the market we choose 1 μF capacitors as C_1 and C_2 .
- Then to achieve the cut-off frequency of 10 Hz the value of the resistors will be 15.9 $\text{k}\Omega$. As it is not available in the market we went with $R_1 = R_2 = 17 \text{ k}\Omega$. This small deviation slightly shifts f_c to around 9.3–9.8 Hz.
- For the operational amplifier we took Op-amp LM339
Designed specifically to detect threshold crossings in slow analog signals like LDR outputs.
Contains four comparators in one IC, reducing component count and simplifying the overall circuit design.

Assumptions made on Physical Design

- Op-amp is powered with a stable single supply
→ LM339 chosen for rail-to-ground compatibility.
- Component tolerances ($\pm 5\%$) do not impact results significantly
→ Butterworth design is robust and tolerant.

Feature 02 – ADC and the 7-Segment Display

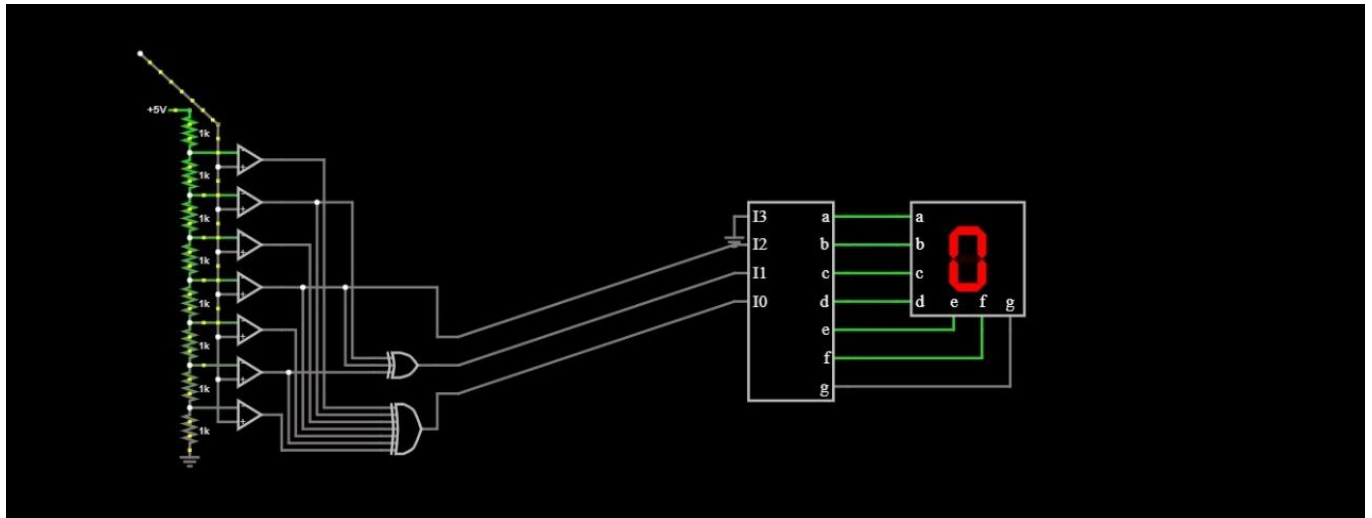
Introduction to the Requirement

Converting the analog voltage from the LDR (Light Dependent Resistor) into a discrete 0–7 light level and displaying that level on a seven-segment display (0 = darkest, 7 = brightest).

Design Approach

Requirement	Design Choice	Assumption
Generate a voltage that varies with ambient light	Use an LDR + resistor voltage divider (sensor voltage increases or decreases monotonically with light)	LDR resistance ranges typically from $\sim 1\text{ k}\Omega$ (bright) to $\sim 100\text{ k}\Omega$ (dark)
Produce evenly spaced reference thresholds	Use a passive resistor ladder with 7 resistors generating 8 taps between 0–VCC	Ladder receives a stable 5 V supply (actual $\approx 4.92\text{ V}$ due to real-world drop)
Detect when sensor voltage crosses each threshold	Use LM339 comparators for reliable analog level comparison	LM339 outputs are open-collector and require pull-ups to 5 V
Generate clean digital signals for logic	Use 1 kΩ pull-up resistors on each comparator output	Comparator outputs will be read by TTL/CMOS logic at 5 V
Convert multiple comparator outputs into a digital value	Use a thermometer-to-binary encoder (priority logic)	Comparators switch in order due to monotonic LDR voltage
Drive a seven-segment display from binary	Use a binary (3-bit) to seven-segment decoder	Display operates on standard 5 V logic levels
Calibrate system for light variations	Add a 10 kΩ trimmer potentiometer in the LDR voltage divide	Final threshold mapping requires field calibration due to LDR tolerance

Falstad Simulation



Mathematical calculations and circuit design

1. Resistor Ladder (Threshold Voltage Generation)

We need 8 light-level steps (0–7).

So we use 7 equal resistors to create 7 equally spaced threshold voltages.

Resistor Ladder Values :

Each resistor: 1 kΩ

Total resistance: $R_{\text{total}} = 7 \times 1\text{k}\Omega = 7\text{k}\Omega$

Ladder Current :

Using supply = 5 V

$I = 5/7000 \approx 0.693 \text{ mA}$

Current is small → low power, stable output nodes.

Threshold Voltages

Equal steps of : $V_{\text{step}} = 5.0 \text{ V}/8 = 0.625 \text{ V}$

So for the each tap the threshold voltage increases by 0.625 Volts.

Each comparator checks whether **Sensor_Voltage** > **V_threshold**.

2. Why 1 kΩ Ladder Resistors & 10 kΩ Pull-Ups

1 kΩ Ladder Resistors :

Give lower output impedance → better comparator accuracy.

Still consume less than 1 mA → acceptable for a small project.

Produce stable, noise-resistant threshold voltages.

10 k Ω Pull-Ups (LM339 outputs) :

LM339 has open-collector outputs → needs pull-ups.

10 k Ω gives:

- Stable logic HIGH.
- Good switching speed for <10 Hz signals.

Physical Design and Assumptions made on it

1. Comparator Section (LM339)

LM339 operates correctly at a 5 V supply.

Input signal to comparators is slow and already filtered, so no high-frequency noise issues.

2. Thermometer Code → Binary Logic

Thermometer output pattern is correctly converted to a 3-bit binary code by logic gates.

LM339 output levels (after pull-ups) are compatible with the logic gate inputs.

The 3-bit binary code drives a standard seven-segment decoder without conflict.

3. Seven-Segment Display

The display accepts standard TTL/CMOS logic levels.

The decoder IC correctly maps binary inputs 0–7 to segment patterns.

Segment current is within safe operating limits for continuous operation.

4. Supply Voltage and Thresholds

The real supply voltage is around 4.92 V, not exactly 5 V, and is used for threshold accuracy.

This small voltage drop only shifts ladder thresholds slightly, without affecting functionality.

The resistor ladder produces evenly spaced thresholds because all resistors have equal values.

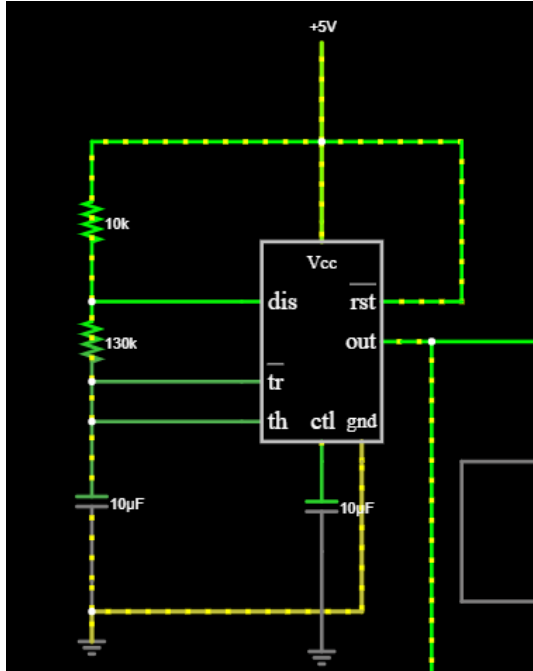
Feature 03 – Stabilizer Circuit

Introduction to the Requirement

The module should block sudden or temporary changes in light intensity.

The output must change only when the light level remains stable for a selected delay, adjustable between 30 s and 300 s. A variable resistor changes the delay, and a switch will later enable/disable this feature. This ensures the displayed light level is steady and not affected by momentary flicker.

1. Clock circuit



The stability detection feature requires a timing mechanism to ensure the light intensity display only updates after the light has remained stable for an adjustable period (30-300 seconds). A **555 Timer IC in astable mode** serves as the clock generator, providing periodic pulses that drive the stability monitoring logic.

555 Timer Astable Multivibrator Configuration

The 555 timer is configured to generate continuous square wave pulses with an adjustable period. The circuit operates in two phases:

Charging Phase (Output HIGH):

Timing capacitor C1 charges through resistors R1 and R2 and when voltage reaches $\frac{2}{3} V_{cc}$, internal comparator triggers a change of state

Discharging Phase (Output LOW):

Capacitor discharges through R2 only via the discharge pin and when voltage drops to $\frac{1}{3} V_{cc}$, cycle restarts again.

555 Timer IC pins and its use

Vcc: Power supply (+5V)

gnd: Ground

rst: Reset pin (connected to Vcc to keep timer running)

out: Output (square wave signal)

dis: Discharge pin

tr: Trigger pin

th: Threshold pin

ctl: Control voltage pin

10kΩ Resistor (R1)

Connected between Vcc and discharge pin.

Provides minimum timing baseline; ensures circuit stability; much smaller than R2_max to allow R2 to dominate timing adjustment.

130kΩ Resistor (R2)

Variable resistor (potentiometer) that adjusts the timing period between 30-300 seconds.
allows user to dial in precise delay (30-300s range) without circuit modification
Simple rotational control.

10μF Capacitor (C1)

Timing capacitor that charges/discharges to create the clock period. Large capacitance value required to generate slow clock signals in seconds range; creates long RC time constant with large resistance

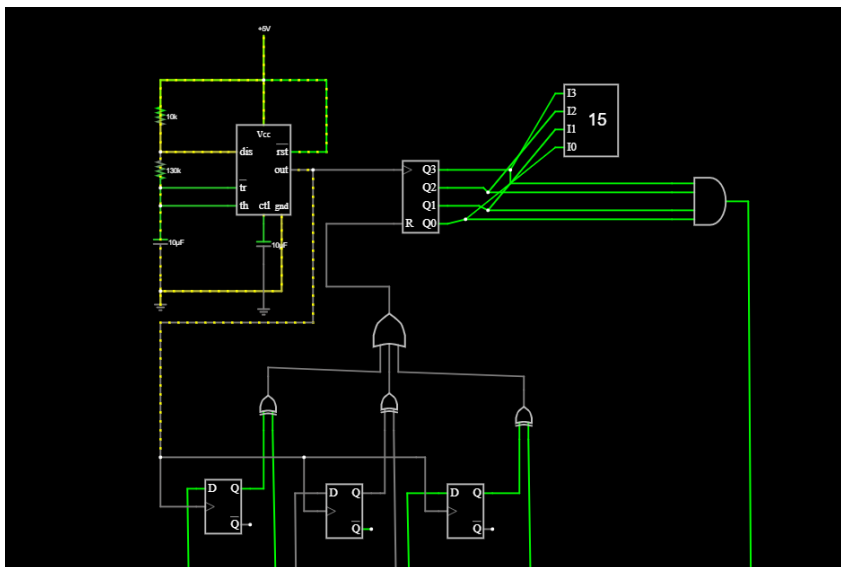
10μF Capacitor (C2)

Connected to control voltage pin for stability/noise filtering. Same value as C1 for simplicity; provides adequate filtering to stabilize internal reference voltage and reduce noise-induced jitter.

Assumptions made during the Physical Design

- Output drives CMOS logic gates; minimal current draw ensures 555 output is not overloaded
- Assumes stable ambient temperature; component values and timing remain relatively constant
- Standard tolerance for resistors and capacitors; actual timing periods may vary slightly from calculated values

2. Update and Reset Circuit



The update signal circuit determines when the main display should refresh with a new stable light intensity reading. This circuit implements a counter-based timing system that counts clock pulses and generates an update signal once the stability period is complete, with reset capability to restart timing when light intensity changes.

4-Bit Binary Counter

The counter receives clock pulses from the 555 timer and increments its count with each pulse. The counter outputs (Q3, Q2, Q1, Q0) represent a 4-bit binary value ranging from 0000 to 1111 (decimal 0-15). The counter includes a Reset (R) pin that clears all outputs back to 0000 when triggered, allowing the timing cycle to restart.

Why 4-bit counter: Counts 15 clock pulses before reaching maximum, providing sufficient time divisions. Combined with adjustable 555 timer period, achieves the required 30-300 second stability detection range.

4-Input AND Gate (Decoder)

Connected to all four counter outputs (Q3, Q2, Q1, Q0), this gate monitors the counter state continuously. The AND gate output goes HIGH only when all counter bits equal 1, detecting the terminal count of 1111 (decimal 15). This signal indicates the stability period has elapsed.

Why AND gate: Simple and efficient method to decode when counter reaches maximum count; requires only one gate to detect the specific count value; provides clean digital signal for triggering update.

D Flip-Flops (Three Cascaded Stages)

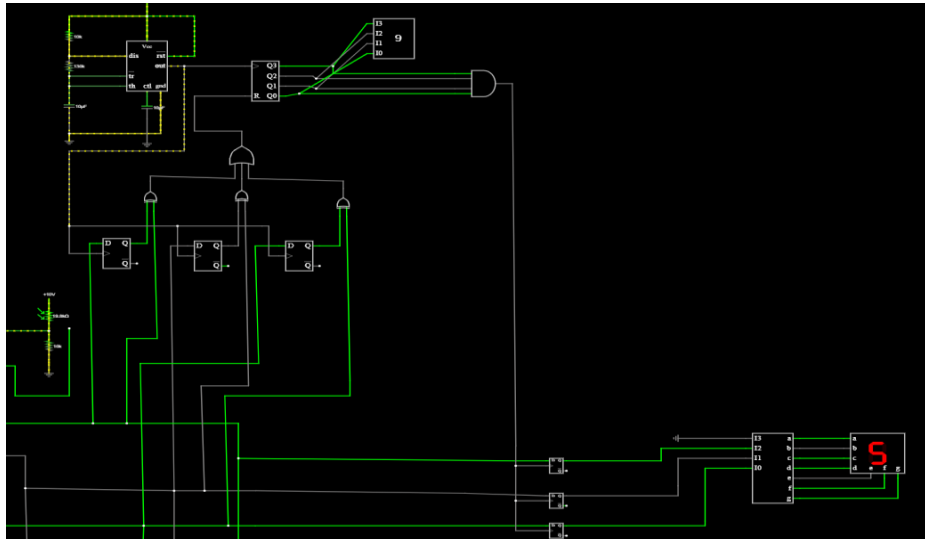
Three D flip-flops arranged in series provide signal synchronization and storage of the current light intensity level. The three D inputs to these flip-flops receive the 3-bit output from the ADC, which represents the digitized light intensity level (0-7). Each flip-flop captures and holds its input value on the clock edge, preventing glitches and ensuring clean state transitions. Hold previous stable reading until new stable value confirmed.

Assumptions made during the Physical Design

- Counter propagation delay negligible compared to clock period
- Flip-flop setup and hold times satisfied by slow clock frequency
- Clean clock signal from 555 timer; no additional debouncing required

3. Switch mechanism

Whole circuit operation of the stabilizer circuit



CASE 1: Light Intensity Remains STABLE (No Change)

- **Initial Operation**
 - ADC converts analog light to 3-bit digital value (I2, I1, I0) representing levels 0-7
 - First D flip-flop set stores current ADC reading
 - Second D flip-flop set holds last displayed stable value
 - 555 timer generates clock pulses every ~2 seconds
- **Comparison Check**
 - Comparator circuit (XOR gates) compares current reading vs. displayed value
 - Both values identical → Comparator output = LOW (no change detected)
 - No reset signal generated
- **Counter Counts Up**
 - 4-bit counter increments with each clock pulse: 0000 → 0001 → ... → 1111
 - Continues counting since no reset signal present
- **Terminal Count Reached**
 - When counter reaches 1111 (count = 15), 4-input AND gate output goes HIGH
 - Indicates: "Light stable for required period"
- **Display Updates**
 - AND gate HIGH output clocks second flip-flop set
 - Second flip-flops latch current stable value from first flip-flops
 - 7-segment decoder converts 3-bit value to display pattern
 - Display shows new stable light level (0-7)
- **Cycle Resets**
 - Counter resets to 0000 after update
 - New stability monitoring cycle begins

CASE 2: Light Intensity CHANGES (Unstable)

- **Change Detected**
ADC outputs new 3-bit value due to light intensity change
First flip-flop set captures new reading
- **Comparison Mismatch**
Comparator detects difference between current reading and displayed value
Comparator output goes HIGH → "Change detected!"
- **Counter Immediately Resets**
HIGH comparator output generates reset signal
Counter clears to 0000, aborting current timing cycle
Counting restarts from zero
- **Display Unchanged**
Second flip-flop set NOT clocked → retains last stable value
Display shows previous stable reading
- **New Cycle Begins**
System must now count full 15 pulses with stable light
If light changes again → counter resets again
If light stabilizes → counts to 15 → updates display

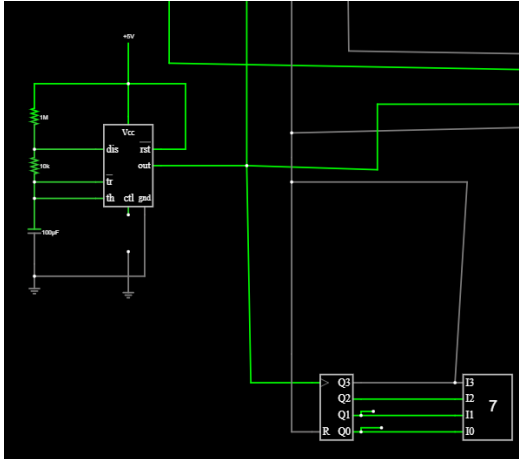
Feature 04 – Averaging the Light Intensity Over a Time Period

Introduction to the Requirement

This feature incorporates an additional seven-segment display to indicate the average light intensity over a specified period, adjustable between 300 and 900 seconds. A variable resistor modifies this period, and a push button resets the average reading. This feature provides a comprehensive understanding of light conditions over an extended period by smoothing out temporary fluctuations and offering an accurate holistic view of the lighting environment.

1. Clock circuit for averaging period

Uses a 555 Timer IC in astable mode to generate sampling clock pulses. The circuit operates identically to Feature 3's clock circuit, but with different component values to achieve longer sampling intervals suitable for the 300-900 second averaging window.



Timing Logic:

Averaging period: 300-900 seconds (by variable R2)

Number of samples: 8 samples (power of 2 for binary division)

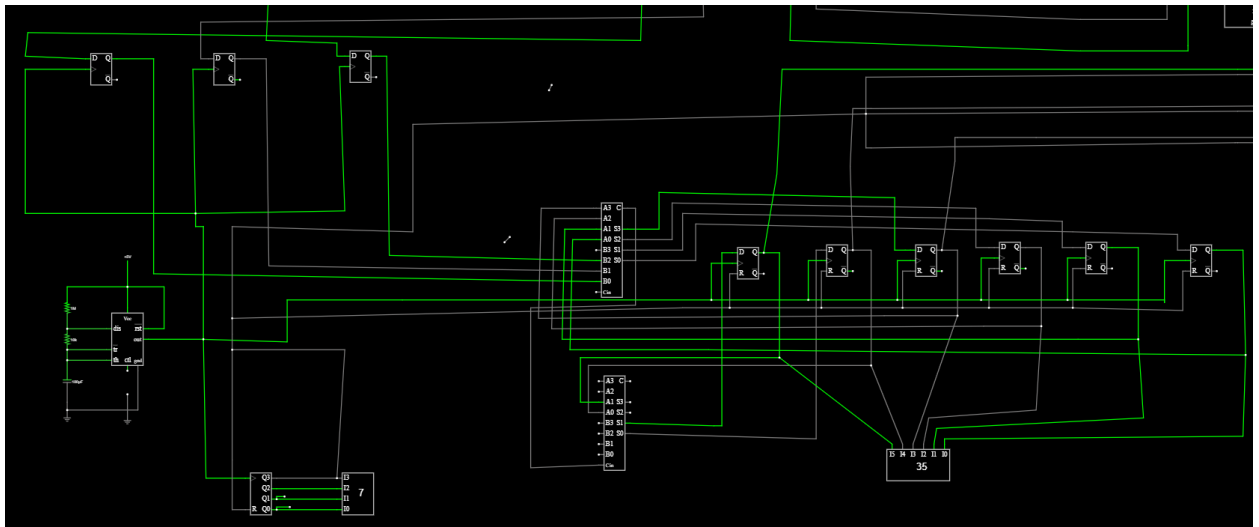
Time per sample: 37.5s (minimum) to 112.5s (maximum)

Why 8 samples: Division by 8 = right shift 3 bits (hardware efficient)

Key Design Decisions and Assumptions

Design Choice	Justification / Assumption
R2 = 500k Ω variable	Provides full 300-900s range through potentiometer adjustment
C1 = 100 μ F	Extended RC time constant for long sampling periods
Output drives CMOS	Minimal current draw; 555 output not overloaded

2. Sample Storage and Accumulation Circuit

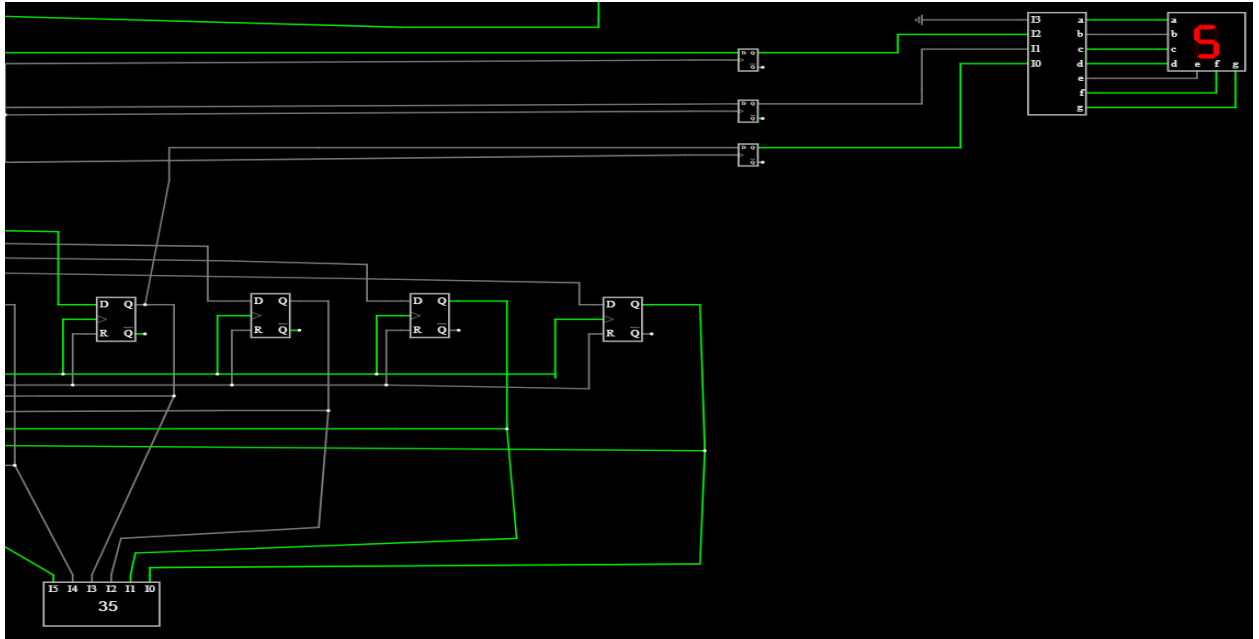


This circuit captures 8 sequential light intensity samples and accumulates them into a running sum.

- 3 Input D Flip-Flops (Sampling Register)**
D inputs: Receive 3-bit ADC output - continuously changing
Clock: 555 timer pulses (every 37.5-112.5s)
Q outputs: Stable 3-bit sample to accumulator circuit
Purpose: Sample-and-hold; captures exact ADC value at sampling instant; holds stable during addition
Why needed: ADC output continuously varies; these flip-flops synchronize sampling to clock intervals and provide stable input to adder.
- Binary Adder ICs (Two ICs)**
Upper IC: Primary 6-bit binary adder
Lower IC: Secondary adder stage or control logic
Purpose: Adds new 3-bit sample (extended to 6 bits as 000XXX) to current 6-bit accumulator value
Output: 6-bit sum = Previous accumulator + Current sample
- 6 D Flip-Flops (Accumulator Register)**
Purpose: Store running sum of all samples collected
Capacity: 6 bits can hold 0-63 (maximum sum = $8 \times 7 = 56$)
Operation: Stores adder output; feeds back as input for next addition
- Sample Counter**
 3-bit binary counter
Range: Counts 000 to 111 (decimal 0-7, representing 8 samples)
Purpose: Tracks number of samples collected
Terminal detection: Counter reaches 111 → triggers averaging
- Multiplexer IC 35**
 Controls data routing and operation mode
During sampling (count 0-6): Routes adder output back to accumulator for continuous accumulation
At count 7 (8th sample): Switches to averaging mode; triggers display update
Control: Manages transition between accumulation and averaging phases

Design Choice	Justification / Assumption
Dual adder ICs	Handle 6-bit addition; cascaded for proper carry propagation
IC 35 multiplexer	Controls data flow; mode selection; single IC solution
Synchronous clocking	All flip-flops share 555 clock; no timing issues

3. Average Calculation and Display Circuit



This stage performs binary division ($\div 8$) and displays the result.

- **Binary Division by 8 (3-Bit Right Shift)**

Logic: Since $8 = 2^3$, dividing by 8 = right shift 3 positions.

6-Bit Accumulator: B5 B4 B3 B2 B1 B0

Division: Upper 3 bits (B5 B4 B3) = average; lower 3 bits (B2 B1 B0) discarded

Example: Sum = 44 = 101100 $\rightarrow$ Average = 101100 $\gg$ 3 = 101 = 5 (decimal)

- **3 Output D Flip-Flops (Average Storage)**

D inputs: Connected to B5, B4, B3 (upper bits of accumulator)

Clock: Triggered when counter reaches 111 (8 samples complete)

Q outputs: Feed to seven-segment decoder/display

Purpose: Latch stable average; prevent display flicker during next sampling cycle

- **Display Update Process**

When Counter = 111: IC 35 detects 8 samples complete

Output flip-flops clock signal activated

Upper 3 bits (B5 B4 B3) latched into output flip-flops

Seven-segment display updates with new average

Accumulator resets to 000000

Counter resets to 000

New cycle begins

- **Seven-Segment Display:** Shows average light level 0-7 continuously until next update (every 300-900 seconds depending on R2 setting).

- **Reset Button Functionality**

Push button clears accumulator and counter to 000

Display retains last valid average

New sampling cycle begins immediately

Reset logic integrated within system control

Design Choice	Justification / Assumption
Division by right-shift	Hardware-efficient; direct wiring B5→D2, B4→D1, B3→D0
Fractional bits discarded	Integer display sufficient; simplifies design
3 output flip-flops	Stable display; no flicker during sampling